

Northeastern University

Principles and Applications of Remote Sensing (Bilingual) Computer Practice Report



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1. Computer experiment content (1): band conversion, radiation correction

1. Experimental content

① **Data preparation:** Download Landsat or Sentinel images with an interval of more than five years between two scenes, and the image area is your hometown or other areas of interest. Requirements: At least one of the downloaded data is new data in recent years; Clear and good-quality images without clouds are needed; You need to try to download images in the same month or quarter; It needs to include 6 types of land use types: cultivated land, forest land, grassland, water body, construction land and unused land. If there are less than 6 types of land features in the study area, please select another area!

② **Radiation calibration:** Find the calibration parameters, select the radiation calibration tool, set the calibration parameters, and execute the radiation calibration

③ **Atmospheric correction:** Open the FLAASH module, input images and parameters, set output parameters, perform atmospheric correction, remove the influence of the atmosphere, and calculate the true reflectivity of the surface.

④ **View the correction results:** select at least three ground features and describe the differences in pixel values of the three ground features in different wavelength bands.

2. Experimental purpose

① Understand the band characteristics of remote sensing images: master the band composition and physical significance of multispectral remote sensing images, and understand the differences in the response of different bands to ground features.

② Master the principles and methods of radiation correction: learn the principles of radiation calibration and atmospheric correction, and master the technical process of eliminating sensor errors and atmospheric interference.

③ Proficient in remote sensing software operation skills: Complete band combination transformation and radiation correction operations through tools such as ENVI or ArcGIS to improve remote sensing data processing capabilities.

④ Cultivate quantitative analysis ability: analyze the influence of radiation correction on remote sensing quantitative inversion (such as vegetation index and surface temperature) by comparing images before and after correction.

3. Experimental principle

1. **Band conversion principle:** Multispectral remote sensing images are composed of multiple bands with different wavelength ranges, and each band reflects different spectral characteristics of ground objects. Band conversion is to combine and operate different

bands (such as addition, subtraction, multiplication, ratio operation, etc.) to highlight the characteristics of the features of interest and suppress the influence of other features or noise. For example, ratio calculation can enhance vegetation information, because the reflectivity of vegetation in the near-infrared band and the red light band is significantly different. Through the ratio calculation of the near-infrared band and the red light band, the difference between vegetation and other ground features can be highlighted.

2. Principle of radiation calibration: As a key link in remote sensing image processing, the main purpose of radiation calibration is to correct radiation errors caused by factors such as sensor response characteristics, atmospheric influence, differences in lighting conditions, etc., and ensure that remote sensing images can accurately reflect the reflection or emission characteristics of ground objects. In the process of receiving the radiation signal and converting it into digital quantized value (DN), the response function of the sensor may show nonlinear characteristics. In order to accurately calibrate this nonlinear response, it is necessary to determine the quantitative relationship between the sensor input radiance (L) and the output DN value. Usually, laboratory calibration or site calibration means are used to obtain data, and then polynomial functions are used for fitting:

$$DN = a_0 + a_1L + a_2L^2 + \dots + a_nL^n$$

Where, a_0 、 a_1 、 a_2 、...、 a_n Is the coefficient determined by calibration. In actual operation, multiple groups of different radiance L and their corresponding DN values will be collected, and these coefficients will be solved by fitting algorithms such as the least square method, and then an accurate radiation calibration model will be constructed to effectively correct the sensor response and provide a reliable data basis for subsequent quantitative analysis.

3. Atmospheric correction principle: The scattering and absorption of electromagnetic waves by the atmosphere seriously interferes with the radiation information of remote sensing images, resulting in radiation errors, and making it difficult for images to truly reflect the reflectivity of ground objects. Therefore, atmospheric correction becomes a necessary step to obtain accurate reflection information of ground objects. Among many atmospheric correction methods, the method based on radiative transfer model is widely used. By carefully considering the complex processes of atmospheric radiation, such as scattering, absorption and reflection, the radiation transfer equation is established to invert the true reflectance of ground objects. A common simplified radiative transfer equation is:

$$L = \frac{\pi L_{path} + \rho E_{solar} T}{\pi - \rho S}$$

Where L represents the radiance received by the sensor; L_{path} It is the brightness of atmospheric radiation, which is the part of radiation scattered by the atmosphere directly into the sensor, and is closely related to the composition and thickness of the atmosphere; E_{solar} It is the incident irradiance of the sun, which depends on factors such as solar altitude angle, geographical location and time; T represents the atmospheric transmittance, which reflects the proportion of radiation that is not absorbed and scattered when it is transmitted in the atmosphere; S is the atmospheric scattering coefficient, which is related to the properties and concentrations of scatterers such as aerosols and molecules in the atmosphere. In practical application, the parameters in the equation are obtained by field measurement, model estimation or reference to relevant data, and then the above equation is solved to invert the true reflectance of ground objects ρ . Thereby eliminating the interference of the atmosphere on the radiation information of remote sensing images, completing the atmospheric correction work, and providing reliable data for subsequent high-precision remote sensing analysis such as ground feature classification and vegetation index calculation.

4. Experimental results and analysis

1. Analysis of results before and after atmospheric correction of remote sensing images in Baoding in 2014:



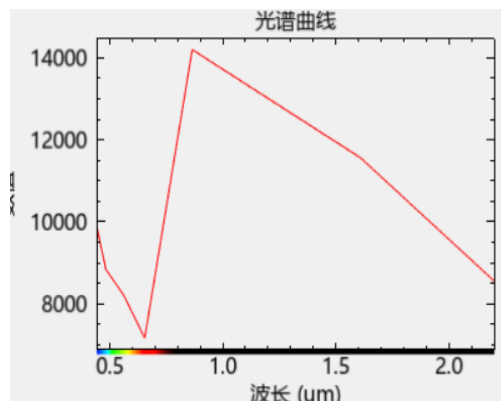
(a) Remote sensing image before atmospheric correction (b) Remote sensing image after atmospheric correction

Before atmospheric correction, remote sensing images are affected by atmospheric scattering and absorption, and the true reflectivity information of ground objects is interfered. The overall clarity and contrast of the images are poor, and the boundaries of ground objects are blurred,

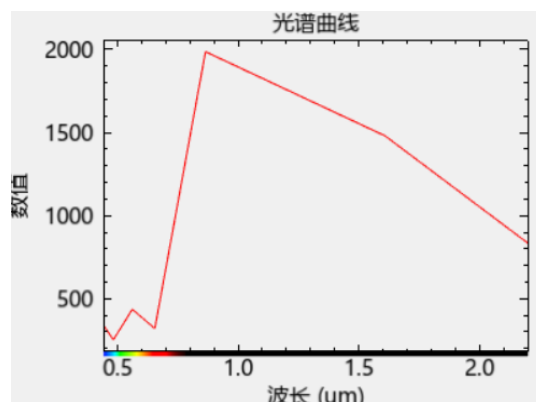
making it difficult to accurately identify and distinguish different types of ground objects. After atmospheric correction, atmospheric interference is effectively removed, the image is clearer, the boundary of ground features is clearer, and the true characteristics of ground features can be better reflected, providing a reliable data basis for subsequent accurate ground feature classification, quantitative inversion and other analysis.

2. Comparison of spectral curves of different ground objects before and after atmospheric correction:

(1) Comparative analysis of forest spectral curves



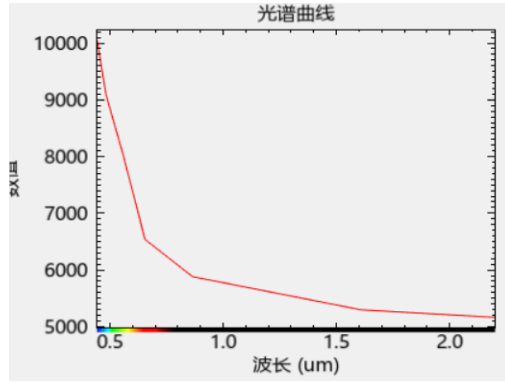
(a) Waveband curves of plants before atmospheric correction



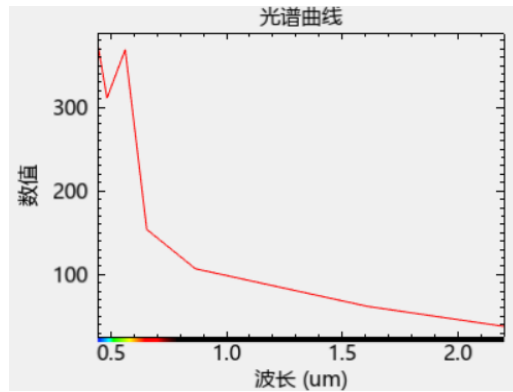
(b) Waveband curves of plants after atmospheric correction

Comparative analysis of forest spectral curves: Before atmospheric correction, the reflectivity values of forest spectral curves in each band are deviated due to the influence of the atmosphere, and the curve fluctuations are inaccurate. After correction, the reflectivity of forest increases significantly in the near-infrared band. This is because atmospheric correction eliminates the attenuation of near-infrared light by the atmosphere, and more truly reflects the high reflection characteristics of vegetation in this band, making the difference between forest and other ground features more obvious in spectral characteristics, which is conducive to extracting and analyzing forest vegetation information in the near-infrared band.

(2) Comparative analysis of water spectral curves before and after atmospheric correction



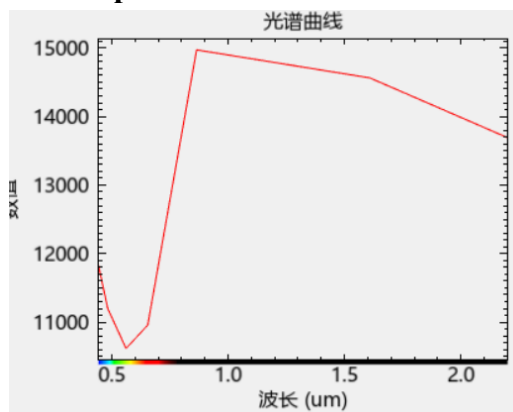
(a) Wavelength band curve of water body before atmospheric correction



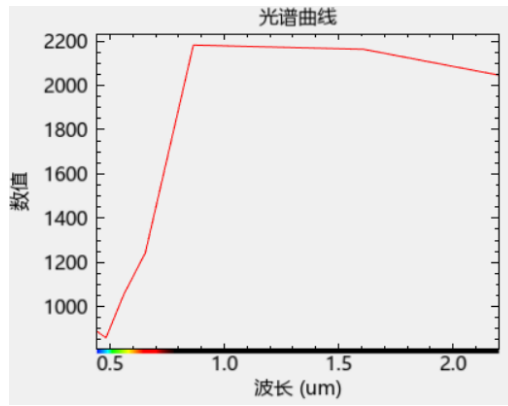
(b) Wavelength band curve of water body after atmospheric correction

Comparative analysis of water spectral curves: Before correction, the water spectral curves were affected by atmospheric scattering, and the reflectivity in the visible light band was high, which could not accurately reflect the light absorption characteristics of the water body. After correction, the reflectivity of the water body in the visible light band decreases, especially in the blue light and green light bands, which conforms to the law of water body absorption of light. The reflectivity in the near-infrared and short-wave infrared bands approaches zero. This change makes the water body in the image The features in the image are more prominent, which is helpful for the accurate extraction and identification of water body information.

(3) Comparative analysis of urban spectral curves before and after atmospheric correction



(a) Band curves of cities before atmospheric correction



(b) Band curves of cities after atmospheric correction

Comparative analysis of urban spectral curves: Before atmospheric correction, urban spectral curves are affected by the atmosphere and have atypical characteristics, making it difficult to accurately distinguish cities from other ground features based on spectral characteristics. After correction, the reflectivity of the city in the visible light band is adjusted, and the spectral difference with the surrounding features is more significant. In the near infrared band, the reflectivity difference between the city and vegetation increases, which is convenient to distinguish urban construction land from vegetation coverage area through spectral analysis.

2. Contents of computer experiments (2): Supervised classification of remote sensing images

1. Experimental content

① Data preparation: high-quality remote sensing images after atmospheric correction, cropped images after radiation correction, the cropped image features should be as rich as possible, and the cropped size should not be less than 50km * 50km. Use multiple (at least three) supervised classification methods for classification (cultivated land, woodland, grassland, water body, building land, unused land), compare the classification accuracy, select the best classification method and map it.

② Sample selection: According to the ground truth data on the remote sensing image, the representative ground object samples are selected as training samples. The samples should cover all types of features of interest in the study area, and have sufficient quantity and distribution uniformity to ensure that the classifier can accurately learn the spectral characteristics of different features.

③ Supervised classification selection: According to the characteristics and research purpose of remote sensing images, select appropriate supervised classifiers, such as maximum likelihood classifier, random forest, etc.

④ Unsupervised classification selection: ISODATA classification method was adopted.

⑤ Accuracy evaluation: Use the reserved ground truth data or independent verification samples to evaluate the accuracy of classification results. Commonly used accuracy evaluation indexes include overall accuracy, classification accuracy, Kappa coefficient, etc.

⑥ Mapping. Map mapping with ArcGIS.

2. Experimental purpose

① Understand the basic principles of supervised classification: master the core concepts of supervised classification (such as training samples, classifiers, feature spaces) and understand the differences between supervised classification and unsupervised classification.

② Master the technical process of supervised classification: learn the complete process from sample selection, classifier training to result post-processing, and be familiar with classification algorithms (such as maximum likelihood, random forest, etc.).

③ Improve the interpretation ability of remote sensing images: Train ground feature recognition and classification skills through practical cases (such as land use/cover classification) to enhance the understanding of spectral features and spatial features.

④ Master the evaluation method of classification accuracy: Learning the evaluation indexes such as confusion matrix, overall accuracy and Kappa coefficient can quantitatively analyze the reliability of classification results.

3. Experimental principle

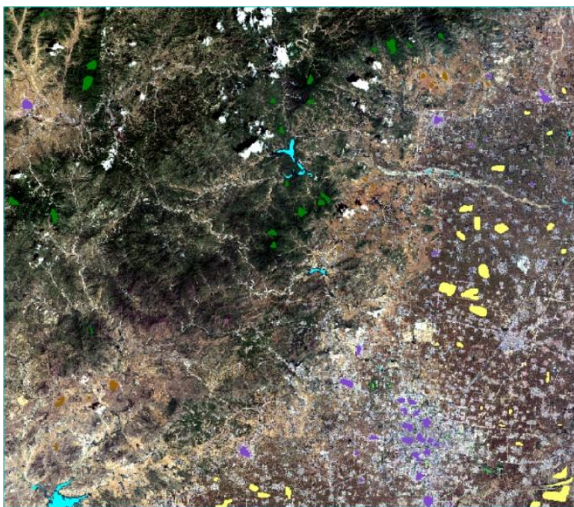
1. Principle of supervised classification: Supervised classification is a classification method based on known samples. Firstly, representative ground feature samples (training samples) are selected from the remote sensing images, which cover all the types of ground features of interest in the study area. Then, using the spectral characteristics of the training samples, the classifier is established by certain classification algorithms (such as maximum likelihood classifier, random forest, etc.). The classifier learns the distribution patterns of different feature categories in the feature space. When encountering unknown pixels, it classifies them into the most similar feature category according to the position of their spectral features in the feature space.
2. Maximum likelihood classifier principle: Assuming that the spectral features of each feature category in the feature space obey the normal distribution, the mean and variance of each category in each band are calculated according to the training samples, so as to obtain the probability distribution model of each category in the feature space. For unknown pixels, the probability of belonging to each category is calculated, and it is classified into the

category with the highest probability.

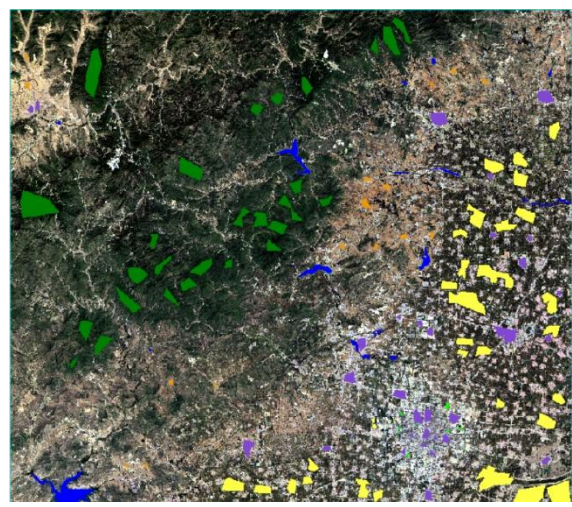
3. Principle of random forest classifier: Random forest is an ensemble learning algorithm based on decision tree. It constructs multiple decision trees by randomly extracting multiple subsets from training samples. In the process of building a decision tree, every time a node is splitted, a part of the features are randomly selected from all the features, and the optimal feature is selected for splitting. Finally, the category of unknown pixels is determined by the voting results of multiple decision trees. This method can effectively reduce the risk of decision tree overfitting and improve the classification accuracy.
4. ISODATA (Iterative Self-Organizing Data Analysis Technology Algorithm) unsupervised classification principle: This method does not need to know the sample category in advance, but aggregates similar pixels into different categories through continuous iterative adjustments based on the spectral similarity between pixels. Firstly, the image pixels are preliminarily clustered according to certain rules (such as specifying the initial number of categories, initial clustering center, etc.). Then, calculate the characteristics of each cluster (such as mean, variance, etc.), merge or split the clusters according to the set thresholds (such as inter-cluster distance threshold, intra-cluster variance threshold, etc.), and continuously optimize the clustering results until the stop conditions are met (such as clustering no longer changes or changes little).

4. Experimental results and analysis

1. Supervise the division of regions of interest before classification



(a) Region of Interest Division in 2014



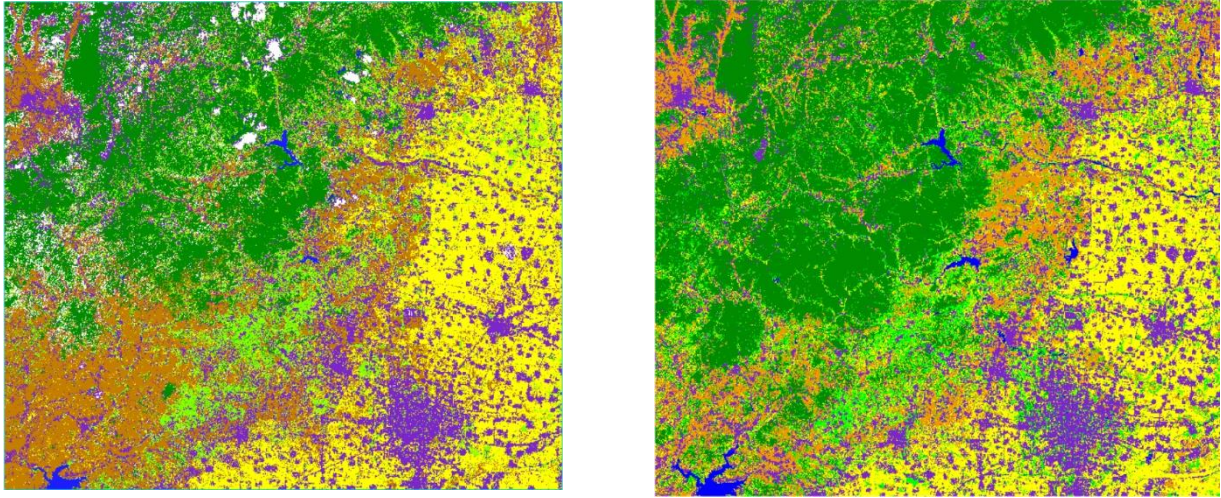
(b) Region of Interest Division in 2022

Among them, the clouds in the 2014 image are distributed on the forest, so they are marked as

forest

2. Results of different classification methods

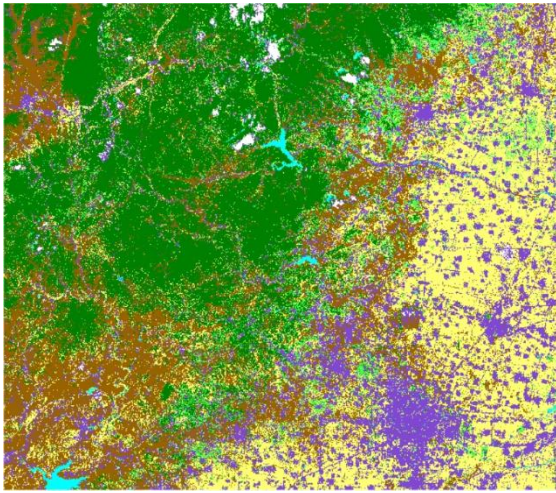
(1) Maximum likelihood classification results



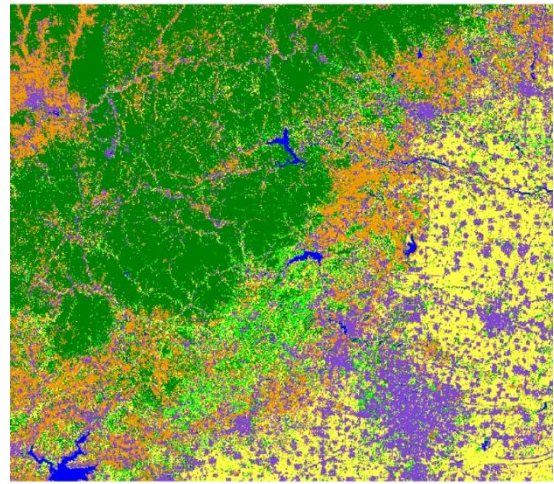
(a) Maximum likelihood classification results in 2014 (b) Maximum likelihood classification results in 2022

Maximum likelihood classification results: In the classification results of 2014 and 2022, it has a good discrimination effect on most feature types. If the boundary of urban areas is relatively clear in the classification map, its distribution range can be accurately presented. However, for some features with similar spectral features, such as bare soil and cultivated land, there is a certain degree of misclassification. This is because the maximum likelihood classifier assumes that the spectral features of features obey normal distribution, and the actual feature spectrum may be complex, resulting in inaccurate classification.

(2) Random forest classification results



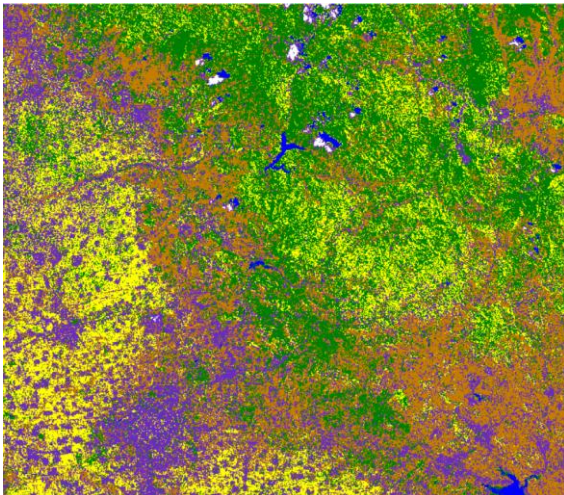
(a) 2014 random forest classification results



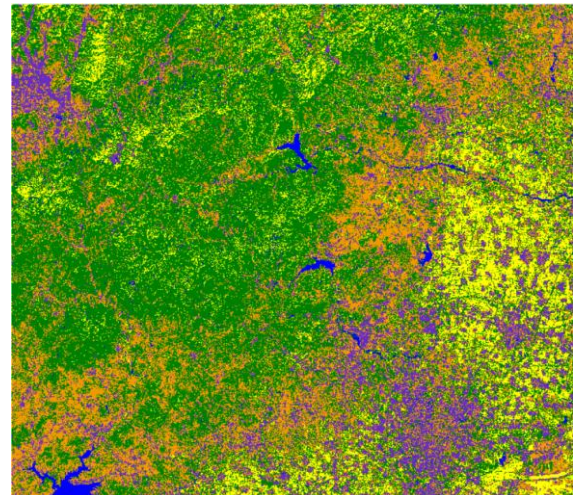
(b) 2022 random forest classification results

Random forest classification results: The overall classification effect is better. Compared with maximum likelihood classification, the ability to distinguish complex features is improved. In the 2022 images, the classification accuracy of forests and grasslands is high, which can reflect their boundaries and distribution in more detail. This is due to the random forest decision tree-based ensemble learning algorithm, which determines categories by voting on multiple decision trees, effectively reducing the risk of overfitting and enhancing the stability and accuracy of classification.

(3) Isodata classification results



(a) 2014 Isodata classification results



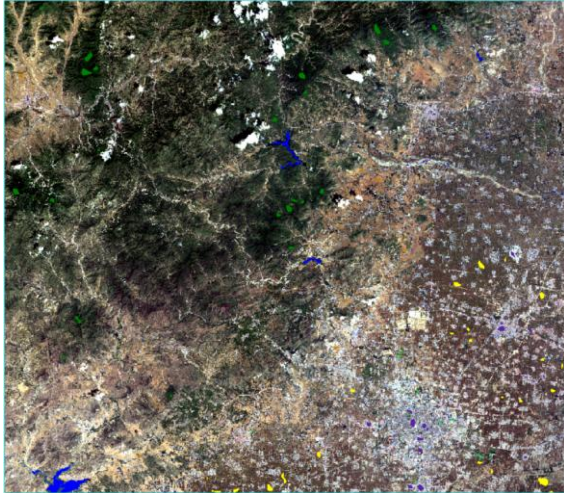
(b) 2022 Isodata classification results

Isodata classification results: In the classification process, the classification accuracy of this method for different features is relatively low due to the lack of prior knowledge. In the classification maps of 2014 and 2022, there are many misclassifications and missed classifications, and the boundaries of ground features are blurred. Especially when distinguishing

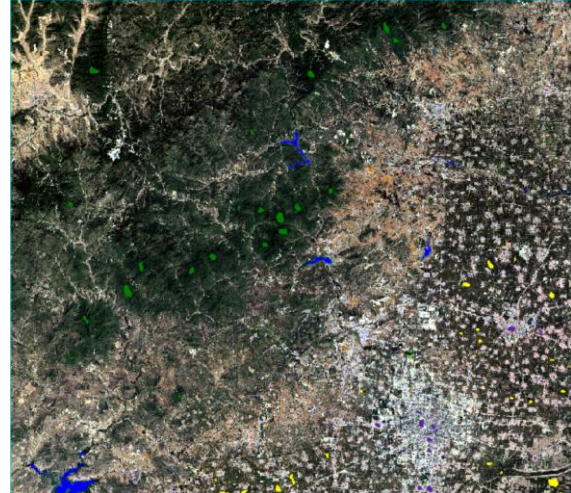
cultivated land from bare soil, urban and other construction land, the effect is poor, and it is difficult to meet the needs of high-precision land use classification.

3. Accuracy evaluation analysis

(1) Re-division of regions of interest for evaluating classification results



(a) Regions of interest for evaluation in 2014



(b) Regions of interest for evaluation in 2022

(2) Maximum likelihood classification accuracy evaluation

Confusion Matrix: C:\Users\15338\Desktop\2014.6.7\class

Overall Accuracy = (39372/41596) 94.6533%

Kappa Coefficient = 0.9314

Class	Ground Truth (Pixels)				
	farm	city	warter	forest	soil
Unclassified	0	0	0	0	0
farm	7454	112	27	1	34
city	220	5296	275	11	15
warter	0	0	12748	0	0
forest	2	0	1	10119	0
soil	267	7	118	54	1488
glass	103	174	469	33	2
cloud	0	156	7	4	0
Total	8046	5745	13645	10222	1539

Class	Ground Truth (Pixels)			Total
	glass	cloud		
Unclassified	0	0		0
farm	32	0	7660	7692
city	43	0	5860	5903
warter	4	0	12752	12756
forest	3	0	10125	10128
soil	48	0	1982	2030
glass	137	0	918	1055
cloud	2	2130	2299	4429
Total	269	2130	41596	43995

Class	Ground Truth (Percent)				
	farm	city	warter	forest	soil
Unclassified	0.00	0.00	0.00	0.00	0.00
farm	92.64	1.95	0.20	0.01	2.21
city	2.73	92.18	2.02	0.11	0.37
warter	0.00	0.00	93.43	0.00	0.00
forest	0.02	0.00	0.01	98.99	0.00
soil	3.32	0.12	0.86	0.53	96.69
glass	1.28	3.03	3.44	0.32	0.13
cloud	0.00	2.72	0.05	0.04	0.00
Total	100.00	100.00	100.00	100.00	100.00

Confusion Matrix: C:\Users\15338\Desktop\2022.6.5\class2

Overall Accuracy = (44988/45921) 97.9682%

Kappa Coefficient = 0.9722

Class	Ground Truth (Pixels)				
	glass	soil	city	forest	warter
Unclassified	0	0	0	0	0
glass	171	0	136	67	13
soil	0	1538	14	0	0
city	15	0	4930	0	0
forest	19	0	0	14574	0
warter	13	0	9	0	16160
farm	16	1	171	4	0
Total	234	1539	5260	14645	16173

Class	Ground Truth (Pixels)			Total
	farm	cloud		
Unclassified	0	0		0
glass	232	0	619	851
soil	119	0	1671	1790
city	104	0	5049	5153
forest	0	0	14593	14593
warter	0	0	16182	16182
farm	7615	0	7807	15422
Total	8070	0	45921	53991

Class	Ground Truth (Percent)				
	glass	soil	city	forest	warter
Unclassified	0.00	0.00	0.00	0.00	0.00
glass	73.08	0.00	2.59	0.46	0.08
soil	0.00	99.94	0.27	0.00	0.00
city	6.41	0.00	93.73	0.00	0.00
forest	8.12	0.00	0.00	99.52	0.00
warter	5.56	0.00	0.17	0.00	99.92
farm	6.84	0.06	3.25	0.03	0.00
Total	100.00	100.00	100.00	100.00	100.00

Class	Ground Truth (Percent)			Total
	glass	cloud		
Unclassified	0.00	0.00		0.00
farm	11.90	0.00	18.42	29.32
city	15.99	0.00	14.09	30.08
warter	0.00	0.00	30.66	30.66
forest	1.12	0.00	24.34	25.46
soil	17.84	0.00	4.76	22.60
glass	50.93	0.00	2.21	53.14
cloud	0.74	100.00	5.53	105.77
Total	100.00	100.00	100.00	100.00

Class	Commission (Percent)	Omission (Percent)	Commission (Pixels)	Omission (Pixels)
farm	2.69	7.36	206/7660	592/8046
city	9.62	7.82	564/5860	449/5745
warter	0.03	6.57	4/12752	897/13645
forest	0.06	1.01	6/10125	103/10222
soil	24.92	3.31	494/1982	51/1539
glass	85.08	49.07	781/918	132/269
cloud	7.35	0.00	169/2299	0/2130

Class	Prod. Acc. (Percent)	User Acc. (Percent)	Prod. Acc. (Pixels)	User Acc. (Pixels)
farm	92.64	97.31	7454/8046	7454/7660
city	92.18	90.38	5296/5745	5296/5860
warter	93.43	99.97	12748/13645	12748/12752
forest	98.99	99.94	10119/10222	10119/10125
soil	96.69	75.08	1488/1539	1488/1982
glass	50.93	14.92	137/269	137/918
cloud	100.00	92.65	2130/2130	2130/2299

Class	Ground Truth (Percent)			Total
	farm	cloud		
Unclassified	0.00	0.00		0.00
glass	2.87	0.00	1.35	4.22
soil	1.47	0.00	3.64	5.11
city	1.29	0.00	10.99	12.28
forest	0.00	0.00	31.78	31.78
warter	0.00	0.00	35.24	35.24
farm	94.36	0.00	17.00	111.36
Total	100.00	100.00	100.00	100.00

Class	Commission (Percent)	Omission (Percent)	Commission (Pixels)	Omission (Pixels)
glass	72.37	26.92	448/619	63/234
soil	7.96	0.06	133/1671	1/1539
city	2.36	6.27	119/5049	330/5260
forest	0.13	0.48	19/14593	71/14645
warter	0.14	0.08	22/16182	13/16173
farm	2.46	5.64	192/7807	455/8070

Class	Prod. Acc. (Percent)	User Acc. (Percent)	Prod. Acc. (Pixels)	User Acc. (Pixels)
glass	73.08	27.63	171/234	171/619
soil	99.94	92.04	1538/1539	1538/1671
city	93.73	97.64	4930/5260	4930/5049
forest	99.52	99.97	14574/14645	14574/14593
warter	99.92	99.86	16160/16173	16160/16182
farm	94.36	97.54	7615/8070	7615/7807

(a) Maximum likelihood accuracy evaluation in 2014 (b) Maximum likelihood accuracy evaluation in 2022

(3) Evaluation of random forest classification accuracy

Confusion Matrix: C:\Users\15338\Desktop\2014.6.7\random

Overall Accuracy = (39916/41596) 95.9612%
Kappa Coefficient = 0.9479

Class	Ground Truth (Pixels)				
Unclassified	fara	city	warter	forest	soil
fara	0	0	0	0	0
fara	7544	113	68	57	42
city	233	5311	146	0	4
warter	0	0	13169	3	0
forest	5	0	1	10154	0
soil	207	25	202	2	1490
glass	57	88	59	6	3
cloud	0	208	0	0	0
Total	8046	5745	13645	10222	1539

Class	Ground Truth (Pixels)		Total
Unclassified	glass	cloud	
fara	0	0	0
fara	36	0	7860
city	36	5	5735
warter	9	0	13181
forest	15	0	10175
soil	50	0	1976
glass	123	0	336
cloud	0	2125	2333
Total	269	2130	41596

Class	Ground Truth (Percent)				
Unclassified	fara	city	warter	forest	soil
fara	0.00	0.00	0.00	0.00	0.00
fara	93.76	1.97	0.50	0.56	2.73
city	2.90	92.45	1.07	0.00	0.26
warter	0.00	0.00	96.51	0.03	0.00
forest	0.06	0.00	0.01	99.33	0.00
soil	2.57	0.44	1.48	0.02	96.82
glass	0.71	1.53	0.43	0.06	0.19
cloud	0.00	3.62	0.00	0.00	0.00
Total	100.00	100.00	100.00	100.00	100.00

Confusion Matrix: C:\Users\15338\Desktop\2022.6.5\random

Overall Accuracy = (45263/45921) 98.5671%
Kappa Coefficient = 0.9804

Class	Ground Truth (Pixels)				
Unclassified	fara	warter	forest	city	soil
fara	0	0	0	0	0
fara	7861	0	9	139	1
warter	1	16168	0	8	0
forest	1	4	14495	5	0
city	140	0	0	5060	7
soil	6	0	0	28	1531
glass	59	1	132	20	0
Total	8070	16173	14645	5260	1539

Class	Ground Truth (Pixels)		Total
Unclassified	glass		
fara	0	0	0
fara	23	8033	8033
warter	19	16207	16207
forest	43	14548	14548
city	1	5208	5208
soil	0	1565	1565
glass	148	360	360
Total	234	45921	45921

Class	Ground Truth (Percent)				
Unclassified	fara	warter	forest	city	soil
fara	0.00	0.00	0.00	0.00	0.00
fara	97.41	0.00	0.06	2.64	0.06
warter	0.04	99.97	0.06	0.15	0.00
forest	0.01	0.02	98.98	0.10	0.00
city	1.73	0.00	0.00	96.20	0.45
soil	0.07	0.00	0.00	0.53	99.48
glass	0.73	0.01	0.90	0.38	0.00
Total	100.00	100.00	100.00	100.00	100.00

Class	Ground Truth (Percent)		Total
Unclassified	glass	cloud	
fara	0.00	0.00	0.00
fara	13.38	0.00	18.30
city	13.38	0.23	13.79
warter	3.35	0.00	31.69
forest	5.58	0.00	24.46
soil	18.59	0.00	4.75
glass	45.72	0.00	0.81
cloud	0.00	99.77	5.61
Total	100.00	100.00	100.00

Class	Commission (Percent)	Omission (Percent)	Commission (Pixels)	Omission (Pixels)
fara	4.02	6.24	316/7860	502/8046
city	7.39	7.55	424/5735	434/5745
warter	0.09	3.49	12/13181	476/13645
forest	0.21	0.67	21/10175	68/10222
soil	24.60	3.18	486/1976	49/1539
glass	63.39	54.28	213/336	146/269
cloud	8.92	0.23	208/2333	5/2130

Class	Prod. Acc. (Percent)	User Acc. (Percent)	Prod. Acc. (Pixels)	User Acc. (Pixels)
fara	93.76	95.98	7544/8046	7544/7860
city	92.45	92.61	5311/5745	5311/5735
warter	96.51	99.91	13169/13645	13169/13181
forest	99.33	99.79	10154/10222	10154/10175
soil	96.82	75.40	1490/1539	1490/1976
glass	45.72	36.61	123/336	123/269
cloud	99.77	91.08	2125/2333	2125/2130

Class	Ground Truth (Percent)		Total
Unclassified	glass		
fara	0.00	0.00	0.00
fara	9.83	17.49	172/8033
warter	8.12	35.29	39/16207
forest	18.38	31.69	53/14548
city	0.43	11.34	148/5208
soil	0.00	3.41	34/1565
glass	63.25	0.78	212/360
Total	100.00	100.00	86/234

Class	Commission (Percent)	Omission (Percent)	Commission (Pixels)	Omission (Pixels)
fara	2.14	2.59	172/8033	209/8070
warter	0.24	0.03	39/16207	5/16173
forest	0.36	1.02	53/14548	150/14645
city	2.84	3.80	148/5208	200/5260
soil	2.17	0.52	34/1565	8/1539
glass	58.89	36.75	212/360	86/234

Class	Prod. Acc. (Percent)	User Acc. (Percent)	Prod. Acc. (Pixels)	User Acc. (Pixels)
fara	97.41	97.86	7861/8070	7861/8033
warter	99.97	99.76	16168/16173	16168/16207
forest	98.98	99.64	14495/14645	14495/14548
city	96.20	97.16	5060/5260	5060/5208
soil	99.48	97.83	1531/1539	1531/1565
glass	63.25	41.11	148/234	148/360

(a) Random forest accuracy evaluation in 2014 (b) Random forest accuracy evaluation in 2022

(4) Isodata classification accuracy evaluation

Confusion Matrix: C:\Users\15338\Desktop\2014.6.7\isola1-15

Overall Accuracy = (33620/41327) 81.3512%
Kappa Coefficient = 0.7609

Class	Ground Truth (Pixels)				
Unclassified	warter	soil	city	fara	cloud
warter	11018	0	0	0	0
soil	259	1378	515	385	0
city	75	159	4866	1042	136
fara	25	0	181	6154	0
cloud	0	0	0	0	1994
forest	2268	36.61	183	465	0
Total	13645	1539	5745	8046	2130

Class	Ground Truth (Pixels)		Total
Unclassified	forest		
warter	11	11029	11029
soil	1	2538	2538
city	24	6302	6302
fara	1976	8336	8336
cloud	0	1994	1994
forest	8210	11128	11128
Total	10222	41327	41327

Class	Ground Truth (Percent)				
Unclassified	warter	soil	city	fara	cloud
warter	0.00	0.00	0.00	0.00	0.00
warter	80.75	0.00	0.00	0.00	0.00
soil	1.90	89.54	8.96	4.78	0.00
city	0.55	10.33	84.70	12.95	6.38
fara	0.18	0.00	3.15	76.49	0.00
cloud	0.00	0.00	0.00	0.00	93.62
forest	16.62	0.13	3.19	5.78	0.00
Total	100.00	100.00	100.00	100.00	100.00

Confusion Matrix: C:\Users\15338\Desktop\2022.6.5\isodata10.10

Overall Accuracy = (36753/45687) 80.4452%
Kappa Coefficient = 0.7342

Class	Ground Truth (Pixels)				
Unclassified	warter	fara	soil	forest	city
warter	16164	0	0	0	64
fara	3	6082	0	4346	1
soil	0	361	841	0	1215
forest	6	1565	0	9878	192
city	0	62	698	0	3788
Total	16173	8070	1539	14645	5260

Class	Ground Truth (Pixels)		Total
Unclassified	forest		
warter	16649	0	16649
fara	10432	0	10432
soil	2417	0	2417
forest	11641	0	11641
city	4548	0	4548
Total	45687	0	45687

Class	Ground Truth (Percent)				
Unclassified	warter	fara	soil	forest	city
warter	0.00	0.00	0.00	0.00	0.00
warter	99.94	0.00	0.00	2.87	1.22
fara	0.02	75.37	0.00	29.68	0.02
soil	0.00	4.47	54.65	0.00	23.10
forest	0.04	17.39	67.45	0.00	3.65
city	0.00	0.77	45.35	0.00	72.02
Total	100.00	100.00	100.00	100.00	100.00

Class	Ground Truth (Percent)	
	forest	Total
Unclassified	0.00	0.00
water	0.11	26.69
soil	0.01	6.14
city	0.23	15.25
fara	19.33	20.17
cloud	0.00	4.82
forest	80.32	26.93
Total	100.00	100.00

Class	Commission (Percent)	Oaission (Percent)	Comaission (Pixels)	Oaission (Pixels)
water	0.10	19.25	11/11029	2627/13645
soil	45.71	10.46	1160/2538	161/1539
city	22.79	15.30	1436/6302	879/5745
fara	26.18	23.51	2182/8336	1892/8046
cloud	0.00	6.38	0/1994	136/2130
forest	26.22	19.68	2918/11128	2012/10222

Class	Prod. Acc. (Percent)	User Acc. (Percent)	Prod. Acc. (Pixels)	User Acc. (Pixels)
water	80.75	99.90	11018/13645	11018/11029
soil	89.54	54.29	1378/1539	1378/2538
city	84.70	77.21	4866/5745	4866/6302
fara	76.49	73.82	6154/8046	6154/8336
cloud	93.62	100.00	1994/2130	1994/1994
forest	80.32	73.78	8210/10222	8210/11128

(a) Isodata accuracy evaluation in 2014

Class	Ground Truth (Percent)	
	forest	Total
Unclassified	0.00	0.00
water	36.44	22.83
fara	5.29	25.48
soil	25.48	9.95
city	9.95	100.00
Total	100.00	

Class	Commission (Percent)	Omission (Percent)	Commission (Pixels)	Omission (Pixels)
water	2.91	0.06	485/16649	9/16173
fara	41.70	24.63	4350/10432	1988/8070
soil	65.20	45.35	1576/2417	698/1539
forest	15.14	32.55	1763/11641	4767/14645
city	16.71	27.98	760/4548	1472/5260

Class	Prod. Acc. (Percent)	User Acc. (Percent)	Prod. Acc. (Pixels)	User Acc. (Pixels)
water	99.94	97.09	16164/16173	16164/16649
fara	75.37	58.30	6082/8070	6082/10432
soil	54.65	34.80	841/1539	841/2417
forest	67.45	84.86	9878/14645	9878/11641
city	72.02	83.29	3788/5260	3788/4548

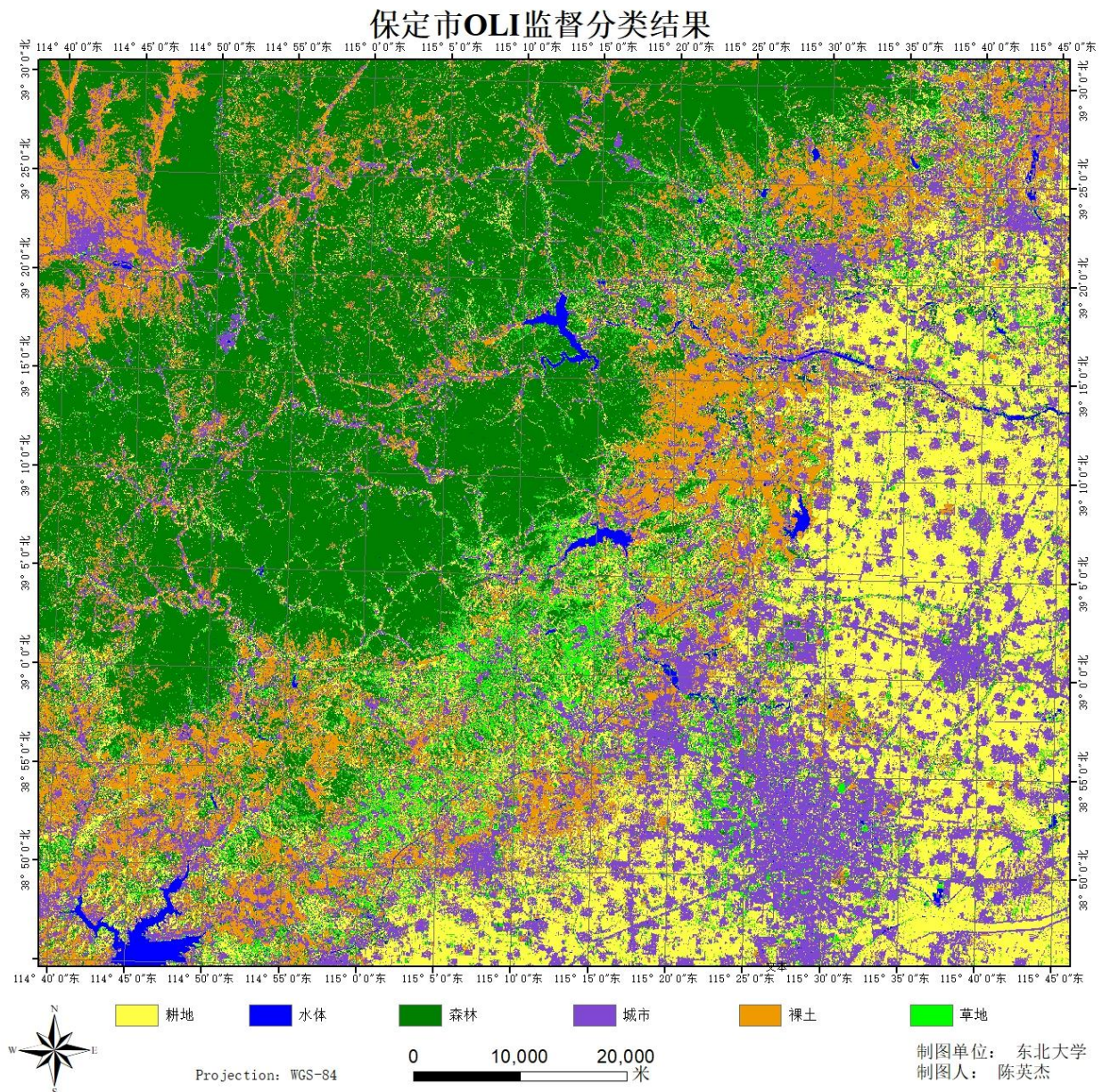
(b) Isodata accuracy evaluation in 2022

Overall accuracy: The overall accuracy of maximum likelihood classification was 94.6533% in 2014 and increased to 97.9682% in 2022; The overall accuracy of random forest classification was 95.9612% in 2014 and reached 98.5671% in 2022; The overall accuracy of Isodata classification was only 81.3512% in 2014 and 80.4452% in 2022. It shows that maximum likelihood and random forest classification methods have high accuracy, and with the passage of time, due to the improvement of image quality and other factors, the classification accuracy is improved, while Isodata classification accuracy is low and unstable.

Kappa coefficient: Maximum likelihood classification Kappa coefficient is 0.9314 in 2014 and 0.9722 in 2022; The Kappa coefficient of random forest classification was 0.9479 in 2014 and 0.9804 in 2022; The Isodata classification has a Kappa coefficient of 0.7609 in 2014 and 0.7342 in 2022. Kappa coefficient reflects the degree of consistency between the classification results and the real situation, which also shows that the maximum likelihood and random forest classification results are in good agreement with the actual situation, while the Isodata classification results are in poor agreement.

Classification accuracy: The classification accuracy of different features is obviously different under different classification methods. For example, in the maximum likelihood classification, the production accuracy of cultivated land in 2014 was 92.64%, and the user accuracy was 97.31%; The production accuracy of random forest classification in 2014 was 92.45%, and the user accuracy was 97.41%; Isodata classification accuracy is low. It shows that maximum likelihood and random forest classification have a good effect on cultivated land classification, and there are many misclassifications and omissions in Isodata classification.

ArcMap mapping results:



3. Content of computer experiment (3): Analysis of land use change

1. Experimental content

① Data preparation and land use classification: Select the best classification method of the last experimental class to classify another image.

② Calculation of land use change: extract the information of land use change and calculate the area.

③ Analysis of land use change: Analyze the transformation of land use types in the two images in the study area, which land type changes to which land type the most, and what are the main driving factors.

1. Experimental purpose

① Master the core methods of land use change detection: understand the principle of multi-temporal remote sensing image change detection, and be familiar with pixel-based and object-oriented change analysis methods.

② Improve multi-temporal data processing capabilities: learn the preprocessing (radiation correction, geometric registration) and data consistency control of two or more remote sensing images.

③ Cultivate spatio-temporal analysis skills: master the technical process of land use dynamic monitoring through change patch extraction, change type classification and driving force analysis.

④ Strengthen practical application ability: Combined with cases such as urban expansion and deforestation, understand the application value of remote sensing in resource management and ecological assessment.

3. Experimental principle

1. Pixel-based land use change detection principle: Compare the spectral characteristics or classification results of the same pixel in two or more remote sensing images to judge whether the pixel has changed. If the difference in spectral values of a pixel exceeds a certain threshold, or if the classification results at different periods belong to different categories, it is considered that the pixel has changed. For example, in the two images, if a pixel is classified as cultivated land in the early stage and construction land in the later stage, it can be determined that the land use change has taken place in the area where the pixel is located.
2. Object-oriented land use change detection principle: Firstly, the remote sensing image is segmented, and the image is divided into several objects with similar spectrum, texture, shape and other features. Then, the characteristics of the corresponding objects in the images of different periods are compared to determine whether the objects have changed or not. This method considers the spatial characteristics of ground features, and can detect land use changes more accurately and reduce the "salt and pepper" phenomenon than the pixel-based method.
3. Principle of land use change analysis: By calculating the transformation matrix between different land use types, the transformation of land use types in two images in the study area is analyzed. The conversion matrix records the area and proportion of each land use type's conversion to other land use types in different periods. From the transformation matrix, it is possible to find out which land class has the most transformations to which land

class. The driving factors of land use change mainly include natural factors (such as climate change, natural disasters, etc.) and human factors (such as urbanization process, agricultural policy adjustment, economic development, etc.). Analyzing these drivers helps to understand the causes and trends of land use change, providing evidence for resource management and planning.

4. Experimental results and analysis

1. Experimental results matrix:

2022年 2014年	耕地	城市	水体	森林	裸土	草地	年总计
耕地	1175.560181	324.8045959	19.8260994	166.1912994	193.2866974	103.0067978	1982.675671
城市	194.539505	978.3386841	23.8491001	66.2885971	96.1766968	43.2486	1402.441183
水体	4.3829999	5.6106	37.5363007	10.7775002	3.4433999	2.5613999	64.3122006
森林	132.4080048	75.9969025	10.3914003	2083.835693	76.7213974	147.691803	2527.045201
裸土	249.9497986	292.5855103	18.7982998	334.6109924	778.3226929	66.6603012	1740.927595
草地	140.9355011	70.8128967	7.7147999	146.675705	45.6831017	75.0140991	486.8361035
年总计	1897.77599	1748.14919	118.1160002	2808.379788	1193.633986	438.183001	8204.237954

2. Results analysis:

Transformation of land use types: From the experimental result matrix, it can be seen that from 2014 to 2022, the transformation area of bare soil to cities was relatively large, reaching 778.3226929, which indicates that a large amount of bare soil was developed as urban construction land in the process of urbanization. The area of cultivated land transformed into cities is 324.8045959, which reflects that urban expansion occupies part of cultivated land resources. At the same time, the change of water area is relatively small, which indicates that the water protection measures in this area are relatively effective.

Driver analysis: Human factors are the main driver. With the acceleration of urbanization, the demand for urban construction has increased, which has prompted a large amount of bare soil and some cultivated land to be converted into urban construction land. Economic development drives urban infrastructure construction and real estate development, and further promotes the transformation of land use types. In addition, policy orientation and population growth demand for urban spatial expansion are also important reasons for land use change. The impact of natural factors is relatively small, but long-term climate change may indirectly affect land use, such as affecting agricultural production and leading to changes in cultivated land use patterns, but it is not clearly reflected in this analysis.